

Checking and Revising Reasoning from Uncertain Evidence

AFOSR FY27 Young Investigator Program White Paper

NOFO: FA955026S0003

Research discipline: Computational Cognition and Machine Intelligence

Program Officer: Dr. Ben Robinson, AFOSR/RESC

Principal Investigator: Xinya Du

Assistant Professor, Department of Computer Science

The University of Texas at Dallas

Email: xinya.du@utdallas.edu

Telephone: +1 (972) 883-2634

Project duration: 36 months

Estimated funding: \$450,000 total (\$150,000 per year)

1. Research Problem and Central Idea

A reasoning system can apply every logical rule correctly and still reach an unsupported conclusion because it misread the evidence. Consider a maintenance report: a sensor failed at 10:02, and a power outage was reported at 10:05. Treating the report time as the outage time would incorrectly rule out the outage as a possible cause. If a later correction places the outage at 9:58, that objection disappears, but timing alone still does not establish causation. The system must distinguish what the report says, what it implies, and what remains unknown.

We ask: *When premises are fallible readings of text, what should a reasoning system check before accepting an answer, and what must it reconsider when a reading changes?* The correct interpretation may be absent from the model’s candidates. Several premises may share the same misreading of a passage. Reusing a check or revising only one argument can therefore save computation while spreading an error.

Our hypothesis is that tracking shared sources, together with logical dependencies, allows a system to target checks and revisions more reliably than tracking logical dependencies alone. We will test whether this reduces unsupported answers at matched computational cost and answer coverage—the fraction of questions answered. The three aims share one argument representation: retain uncertain interpretations, choose checks, and revise conclusions.

Foundation and contribution. The PI’s work on structured reasoning and temporal relation extraction provides methods for representing evidence [4, 6]. MEQA supplies event-centered questions and explanations [7]; FG-PRM provides experience identifying errors within mathematical reasoning steps [8]. Search Wisely studies when additional retrieval is needed [12]. Extending these methods to checking and revising uncertain interpretations is proposed work. Solver-assisted reasoning, truth maintenance, and selective generation provide established foundations [1, 5, 9, 10]. Our research concerns the reliability-cost tradeoff when the premises supplied to those methods may be incomplete or jointly misinterpreted.

2. Research Aims and Technical Approach

Aim 1: Represent what the evidence does and does not establish

Each argument will record source passages, their candidate interpretations, explicit assumptions, and the rules connecting premises to conclusions. We will begin with finite arguments about entities, event order, and fixed conditional rules. Uncertain entity matches and event times will be premises to check, rather than unquestioned inference rules. In the example, the representation will distinguish the time an outage occurred from the time it was reported.

A language model will propose a bounded set of joint interpretations: internally compatible readings of the relevant passages. Separate checks will assess inferences and textual support. Rule checks

will be exact within the specified formal language; evidence checks will be learned and empirically validated. A fixed retention rule, chosen on development data, will determine which interpretations remain plausible. For acceptance, the same answer must have a checked derivation under every retained joint interpretation, with no unresolved evidence check needed by those derivations. An empty candidate set, incompatible answers, or a retained reading with no derivation will require further checking or abstention. Assumption-dependent possibilities will be reported separately.

This criterion has a limited but useful guarantee: with sound rules, an accepted answer follows under the correct joint reading if that reading remains in the candidate set. Rule soundness alone does not establish that the evidence was understood. We will study how candidate generation and pruning trade computational cost against omission of the correct reading, including cases where several premises fail together. Controlled examples and human annotations will measure these omissions independently of the checker. The reported answer and explanation will be derived from the checked arguments; this is a system-level account of support, not a claim to reveal the language model’s internal computations.

Aim 2: Spend checking effort where it can change the decision

In the example, clarifying whether 10:05 is a report time may resolve several premises at once. Separate checks may waste effort; joint acceptance may repeat a mistake. We will compare individual-premise checking with checks that examine a shared passage and its competing interpretations together.

Cost-sensitive testing and model-based diagnosis already study how to choose informative tests [2, 3]. Sequential anomaly search also shows why the cost of switching what is inspected belongs in the decision objective [11]. We will account for measured costs of checking and loading source context when deciding whether to continue examining a passage or move to another. Our setting additionally requires logical support for an answer and allows missing interpretations; the guarantees for anomaly search do not transfer automatically.

We will begin with small acyclic argument graphs and finitely many checks. Each evidence check returns supported, contradicted, or unresolved. Policies will choose the next check using its cost, its estimated effect on the acceptance decision, and the premises that share its source. Learned scores will prioritize checks, not certify their outcomes. The objective is to maximize supported-answer coverage under a budget and an accepted-answer error constraint. An unresolved result will identify the premise or missing check that prevents acceptance.

The formal study will first use specified joint models of check outcomes, permitting correlation among premises from the same source. Dynamic programming over finite checking histories will provide exact reference policies for small cases. We will examine how the number of retained readings and the size of shared-source groups affect this computation, then compare tractable policies with those references. A key question is when grouping checks saves cost and when it hides a decisive alternative. Perturbing the assumed outcome model and removing the correct interpretation will expose failures

of policies optimized under an incomplete model. Document experiments will test the limits of these analyses.

Reliability will be assessed for the complete acceptance procedure. After fixing the policy on development data, independent validation cases will support finite-sample bounds on the unsupported fraction of accepted answers. This uses established statistical tools; correlations inside an argument are not treated as independent validation samples. New document sources will test transfer beyond the validation distribution. We will report coverage and uncertainty intervals.

Aim 3: Reopen the right evidence when an interpretation changes

Classical truth maintenance records which conclusions depend on which assumptions [1]. A source correction can also change premises with no inference edge between them. Discovering that a report uses reporting times rather than event times may require revisiting several timestamps, including ones in otherwise unrelated arguments. We will therefore track both inference dependencies and shared interpretive context.

The central experiment will compare three update procedures: revise only logical descendants of the corrected premise; also recheck premises sharing its source context; or recompute every interpretation and argument. A correction will reopen the affected source group before propagating changes through inference links. Conclusions with independent valid support can be preserved. In the outage example, removing the temporal objection leaves causation unresolved unless other evidence supports it. Every proposed repair must again pass the acceptance procedure; invented premises or weakened rules will not count as repairs.

We will analyze sufficient conditions for local revision to agree with full recomputation under a fixed interpretation procedure, including completeness of the recorded context dependencies. Equivalence for fixed premises is an established baseline. The research question is how broadly to reopen interpretations when those dependencies are uncertain. We will compare passage-level grouping with wider document-level rechecking, measure failures caused by missing links, and study when conservative revision erases the computational savings. Checker-guided refinement, including solver feedback as in Logic-LM [10], will be the initial repair mechanism. A trained repair proposer will be a follow-on experiment if targeted feedback provides a clear advantage. Explicit limits on checks and repair attempts ensure termination; reaching a limit returns unresolved, not successful repair.

3. Evaluation and Measures of Success

Controlled problems will have known facts and valid derivations, with variation in reasoning depth, alternative support, shared-source errors, missing interpretations, and evidence corrections. Independently checked text versions will separate reading errors from inference errors. MEQA will provide a document testbed, extended with human step-level support annotations and corrected-evidence cases.

Ground truth will come from the controlled task specifications and independent human review, never the system’s own checkers. Held-out rule combinations and document sources will test generalization. Questions outside the rule library will count as unresolved rather than being discarded.

Baselines will include direct language-model reasoning, FG-PRM step scoring, solver-assisted reasoning with checker feedback, selective generation, and truth maintenance with uniform or exhaustive evidence checking. All systems will receive the same evidence and be asked to disclose support in the same step format. Missing or unassessable support will not count as a verified derivation. We will match the base model and total computation, including generation, checking, and revision; compare error at matched coverage; and report reliability-cost curves. Ablations will separate logical dependencies, shared-source grouping, and adaptive check selection.

The primary outcome is the fraction of accepted answers whose disclosed support contains an invalid inference or a premise unjustified by the evidence. We will also report this fraction among *correct* accepted answers, using cases where an invalid step can accidentally yield the right answer. Other measures are answer accuracy and coverage, interpretation accuracy, omission of correct readings, and repair success and cost. Evidence interventions will test whether conclusions change when necessary support changes, with alternative valid derivations scored separately. Success requires fewer unsupported answers at comparable coverage and cost, and cheaper correct updates than full recomputation. We will report cases where shared-source checking adds cost without improving reliability.

4. Potential Impact, Team, and Development Plan

Potential impact on DAF and DoW capabilities. Maintenance and anomaly investigations require combining reports that may be incomplete, ambiguous, or later corrected. These methods could help analysts distinguish supported explanations from possibilities, locate supporting reports and assumptions, and identify the next check needed. When a timestamp or equipment identifier is corrected, the system would provide an auditable account of which conclusions changed and why. This basic research would establish when such updates can safely reuse earlier work and when broader review is necessary. It directly addresses the Computational Cognition and Machine Intelligence portfolio’s interests in interpretable foundations, uncertainty, verification, and repair of faulty arguments.

PI Xinya Du will lead the work at UT Dallas with graduate and undergraduate research students. The estimated three-year budget is \$450,000, including institutional indirect costs, for personnel, computation, annotation, and dissemination. No external collaborators or subawards are planned.

Year 1 will build a small end-to-end system covering all three aims, establish the rule and interpretation specifications, and test it on controlled ambiguities and corrections. **Year 2** will develop check-selection policies, analyze restricted cases, and evaluate shared-source errors on annotated documents. **Year 3** will refine revision policies and test reliability and cost across new sources. Deliverables include reproducible code, checker specifications, annotated data, publications, and presentations.

References

- [1] J. de Kleer. An assumption-based TMS. *Artificial Intelligence*, 28(2):127–162, 1986. doi: 10.1016/0004-3702(86)90080-9.
- [2] J. de Kleer and B. C. Williams. Diagnosing multiple faults. *Artificial Intelligence*, 32(1):97–130, 1987. doi: 10.1016/0004-3702(87)90063-4.
- [3] R. Greiner, R. Hayward, M. Jankowska, and M. Molloy. Finding optimal satisficing strategies for and-or trees. *Artificial Intelligence*, 170(1):19–58, 2006. doi: 10.1016/j.artint.2005.09.002.
- [4] X. Jin, H. Wen, X. Du, and H. Ji. Toward consistent and informative event-event temporal relation extraction. In *Proceedings of the First Workshop on Matching From Unstructured and Structured Data*, pages 23–32, 2023. doi: 10.18653/v1/2023.matching-1.3.
- [5] M. Lee, K. Kim, T. Kim, and S. Park. Selective generation for controllable language models. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL <https://arxiv.org/abs/2307.09254>.
- [6] R. Li and X. Du. Leveraging structured information for explainable multi-hop question answering and reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 6779–6789, 2023. doi: 10.18653/v1/2023.findings-emnlp.452.
- [7] R. Li, Z. Wang, S. Q. Tran, L. Xia, and X. Du. MEQA: A benchmark for multi-hop event-centric question answering with explanations. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL <https://openreview.net/forum?id=snNuvAQxB>.
- [8] R. Li, Z. Luo, and X. Du. FG-PRM: Fine-grained hallucination detection and mitigation in language model mathematical reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 4247–4278, 2025. doi: 10.18653/v1/2025.findings-emnlp.228.
- [9] Q. Lyu, S. Havaladar, A. Stein, L. Zhang, D. Rao, E. Wong, M. Apidianaki, and C. Callison-Burch. Faithful chain-of-thought reasoning. In *Proceedings of IJCNLP-AACL 2023*, pages 305–329, 2023. doi: 10.18653/v1/2023.ijcnlp-main.20.
- [10] L. Pan, A. Albalak, X. Wang, and W. Wang. Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 3806–3824, 2023. doi: 10.18653/v1/2023.findings-emnlp.248.
- [11] M. Ubl, B. D. Robinson, and M. T. Hale. Anomaly search over many sequences with switching costs. *arXiv preprint arXiv:2303.09647*, 2023. URL <https://arxiv.org/abs/2303.09647>.
- [12] P. Wu, M. Zhang, X. Zhang, X. Du, and Z. Z. Chen. Search wisely: Mitigating sub-optimal agentic searches by reducing uncertainty. *arXiv preprint arXiv:2505.17281*, 2025. URL <https://arxiv.org/abs/2505.17281>. Accepted to EMNLP 2025.